Linear Algebra and Its Applications

Section 5.7: Applications to Differential Equations

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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5.7 Applications to differential equations

For continuous time the rule is $x'(t) = Ax(t)$. If $Av = \lambda v$, then $ve^{\lambda t}$ is a solution because its derivative is $\lambda ve^{\lambda t} = Ave^{\lambda t}$. For a real eigenbasis, expand the initial vector in that basis and replace each coefficient's evolution by $e^{\lambda t}$.

Complex eigenvalues. If $\lambda = a + bi$ and $v = p + iq$ with p, q real and $b > 0$, a conjugate pair of complex eigenfunctions is $ve^{(a+bi)t}$ and $\overline{v}e^{(a-bi)t}$. A real basis for the same solution space is

$$r_1(t) = e^{at}(p\cos bt - q\sin bt), \qquad r_2(t) = e^{at}(p\sin bt + q\cos bt).$$

This follows by multiplying out Euler's identity $e^{ibt} = \cos bt + i\sin bt$ and taking real and imaginary parts. For real initial data use arbitrary real coefficients of r_1, r_2. The sign of the real part a, rather than $|\lambda|$, decides continuous-time decay or growth.

Practice problem 5.7.P1: Diagonalize over the complex numbers

The three given eigenvalues $-.5, .2+.3i, .2-.3i$ are distinct. Their eigenvectors are consequently linearly independent over $\mathbb{C}$. The matrix $P = [v_1\ v_2\ v_3]$ is invertible and $A = P\,\mathrm{diag}(-.5, .2+.3i, .2-.3i)P^{-1}$.

Practice problem 5.7.P2: Convert the complex general solution to a real one

The complex general solution is $x = c_1v_1e^{-.5t} + c_2v_2e^{(.2+.3i)t} + c_3v_3e^{(.2-.3i)t}$, with complex constants. For real solutions use $v_1 = (1, -2, 1)^T$, $p = (1, 0, 2)^T$, and $q = (2, 4, 0)^T$. The real general solution is

$$x = d_1e^{-.5t}(1, -2, 1)^T + d_2r_1(t) + d_3r_2(t),$$

where r_1, r_2 are the formulas above with $a = .2, b = .3$ and all d_j real. This gives every real solution, since at zero the vectors v_1, p, q are linearly independent.

Practice problem 5.7.P3: Describe the stable line and unstable spiral plane

On the line spanned by v_1 trajectories approach zero like $e^{-.5t}$. In the plane spanned by p, q, they spiral outward with exponential factor $e^{.2t}$. A typical three-dimensional trajectory has both components: its stable-line component decays while its rotating planar component expands. The origin is a spiral saddle, not an attractor.

Exercise 5.7.1: Determine the constants from the initial position

Write $x(0) = c_1v_1 + c_2v_2$. In coordinates this is

$$\begin{bmatrix} -3 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -6 \\ 1 \end{bmatrix}.$$

Solving the two scalar equations gives $c_1 = \frac{5}{2}$, $c_2 = -\frac{3}{2}$. Therefore

$$x(t) = \frac{5}{2}e^{4t}\begin{bmatrix} -3 \\ 1 \end{bmatrix} + \left(-\frac{3}{2}\right)e^{2t}\begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

At $t = 0$ the coefficients reproduce the specified vector. Differentiating each term and using $Av_j = \lambda_jv_j$ verifies $x' = Ax$.

Exercise 5.7.2: Determine the constants from the initial position

Write $x(0) = c_1v_1 + c_2v_2$. In coordinates this is

$$\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Solving the two scalar equations gives $c_1 = \frac{1}{2}$, $c_2 = \frac{5}{2}$. Therefore

$$x(t) = \frac{1}{2}e^{-3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \left(\frac{5}{2}\right) e^{-1t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

At $t = 0$ the coefficients reproduce the specified vector. Differentiating each term and using $Av_j = \lambda_j v_j$ verifies $x' = Ax$.

Exercise 5.7.3: Solve, classify, and identify the principal directions

The coefficient matrix is $A = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$. Factoring its characteristic polynomial gives

$$(\lambda - 1)(\lambda + 1) = 0.$$

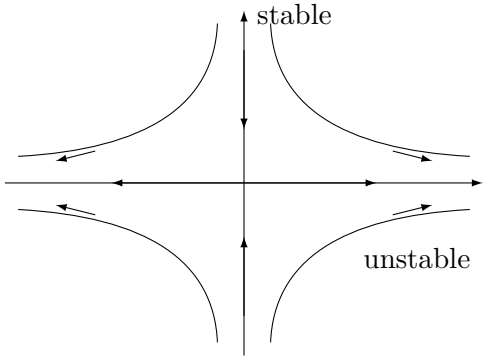
Its two eigenpairs are

$$\lambda_1 = -1,\ v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}; \quad \lambda_2 = 1,\ v_2 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}.$$

Solving $c_1v_1 + c_2v_2 = (3, 2)^T$ gives $(c_1, c_2) = (\frac{9}{2}, -\frac{5}{2})$. Hence the required initial-value solution is

$$x(t) = \left(\frac{9}{2}\right) e^{-1t} \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \left(-\frac{5}{2}\right) e^{1t} \begin{bmatrix} -3 \\ 1 \end{bmatrix}.$$

Both the initial value and $x' = Ax$ check by substitution. The origin is an **saddle point**. The negative eigenvalue's eigenvector v_1 gives the attracting direction; the positive eigenvalue's v_2 gives the repelling direction. The schematic sketch is in these eigen-coordinates: vertical is stable, horizontal unstable. Transforming it by $[v_2\ v_1]$ gives the corresponding picture in the original plane.



Exercise 5.7.4: Solve, classify, and identify the principal directions

The coefficient matrix is $A = \begin{bmatrix} -2 & -5 \\ 1 & 4 \end{bmatrix}$. Factoring its characteristic polynomial gives

$$(\lambda - 3)(\lambda + 1) = 0.$$

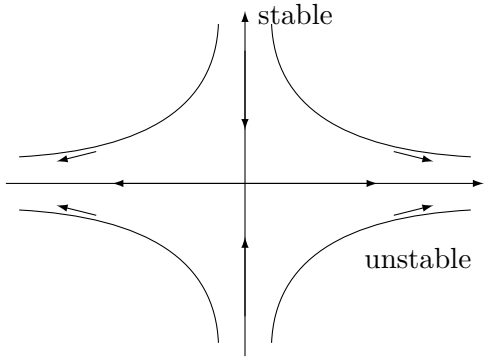
Its two eigenpairs are

$$\lambda_1 = -1,\ v_1 = \begin{bmatrix} -5 \\ 1 \end{bmatrix}; \quad \lambda_2 = 3,\ v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Solving $c_1 v_1 + c_2 v_2 = (3, 2)^T$ gives $(c_1, c_2) = (-\frac{5}{4}, \frac{13}{4})$. Hence the required initial-value solution is

$$x(t) = \left(-\frac{5}{4}\right) e^{-1t} \begin{bmatrix} -5 \\ 1 \end{bmatrix} + \left(\frac{13}{4}\right) e^{3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Both the initial value and $x' = Ax$ check by substitution. The origin is an **saddle point**. The negative eigenvalue's eigenvector v_1 gives the attracting direction; the positive eigenvalue's v_2 gives the repelling direction. The schematic sketch is in these eigen-coordinates: vertical is stable, horizontal unstable. Transforming it by $[v_2\ v_1]$ gives the corresponding picture in the original plane.



Exercise 5.7.5: Solve, classify, and identify the principal directions

The coefficient matrix is $A = \begin{bmatrix} 7 & -1 \\ 3 & 3 \end{bmatrix}$. Factoring its characteristic polynomial gives

$$(\lambda - 6)(\lambda - 4) = 0.$$

Its two eigenpairs are

$$\lambda_1 = 4,\ v_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}; \quad \lambda_2 = 6,\ v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Solving $c_1 v_1 + c_2 v_2 = (3, 2)^T$ gives $(c_1, c_2) = (-\frac{1}{2}, \frac{7}{2})$. Hence the required initial-value solution is

$$x(t) = \left(-\frac{1}{2}\right) e^{4t} \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \left(\frac{7}{2}\right) e^{6t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Both the initial value and $x' = Ax$ check by substitution. The origin is an **repeller**. Both exponential factors grow; the greater eigenvalue 6 gives the fastest repulsion, along $(1, 1)^T$. There is no attracting eigendirection.

Exercise 5.7.6: Solve, classify, and identify the principal directions

The coefficient matrix is $A = \begin{bmatrix} 1 & -2 \\ 3 & -4 \end{bmatrix}$. Factoring its characteristic polynomial gives

$$(\lambda + 1)(\lambda + 2) = 0.$$

Its two eigenpairs are

$$\lambda_1 = -2,\ v_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}; \quad \lambda_2 = -1,\ v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Solving $c_1 v_1 + c_2 v_2 = (3, 2)^T$ gives $(c_1, c_2) = (-1, 5)$. Hence the required initial-value solution is

$$x(t) = (-1)\, e^{-2t} \begin{bmatrix} 2 \\ 3 \end{bmatrix} + (5)\, e^{-1t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Both the initial value and $x' = Ax$ check by substitution. The origin is an **attractor**. Both exponents are negative. The fastest attraction is along $(2, 3)^T$, for eigenvalue -2; the other mode decays more slowly like e^{-t}. There is no repelling eigendirection.

Exercise 5.7.7: A constant eigenbasis decouples the differential equations

Choose the eigenvectors from Exercise 5.7.5 as columns:

$$P = \begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 4 & 0 \\ 0 & 6 \end{bmatrix}.$$

Direct multiplication gives $AP = PD$, and $\det P = -2 \neq 0$. Put $x(t) = Py(t)$. Since P is constant, $x' = Py'$, so

$$Py' = APy = PDy, \qquad y' = Dy.$$

The last equality uses multiplication by P^{-1}. Thus $y_1' = 4y_1$ and $y_2' = 6y_2$, with no cross-coupling. Their scalar solutions are exponentials, and $x = Py$ converts back.

Exercise 5.7.8: A constant eigenbasis decouples the differential equations

Choose the eigenvectors from Exercise 5.7.6 as columns:

$$P = \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}.$$

Direct multiplication gives $AP = PD$, and $\det P = -1 \neq 0$. Put $x(t) = Py(t)$. Since P is constant, $x' = Py'$, so

$$Py' = APy = PDy, \qquad y' = Dy.$$

The last equality uses multiplication by P^{-1}. Thus $y_1' = -2y_1$ and $y_2' = -1y_2$, with no cross-coupling. Their scalar solutions are exponentials, and $x = Py$ converts back.

Exercise 5.7.9: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -3 & 2 \\ -1 & -1 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 + 4\lambda + 5.$$

A positive-imaginary eigenpair is

$$\lambda = -2 + i, \qquad v = \begin{bmatrix} 1 - i \\ 1 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(-2+i)t} + c_2 \overline{v} e^{(-2-i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{-2t}\big[p\cos(1t) - q\sin(1t)\big],$$
$$r_2 = e^{-2t}\big[p\sin(1t) + q\cos(1t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

The sine and cosine produce elliptical rotation in the original plane, and the negative real part produces decay. Every nonzero trajectory spirals inward to the origin.

Exercise 5.7.10: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 - 4\lambda + 5.$$

A positive-imaginary eigenpair is

$$\lambda = 2 + i, \qquad v = \begin{bmatrix} -1 - i \\ 2 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(2+i)t} + c_2 \overline{v} e^{(2-i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{2t}\big[p\cos(1t) - q\sin(1t)\big],$$
$$r_2 = e^{2t}\big[p\sin(1t) + q\cos(1t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

The sine and cosine produce elliptical rotation, while the positive real part expands its size. Nonzero trajectories spiral outward from the origin.

Exercise 5.7.11: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -3 & -9 \\ 2 & 3 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 + 9.$$

A positive-imaginary eigenpair is

$$\lambda = 3i, \qquad v = \begin{bmatrix} -3 + 3i \\ 2 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} -3 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} 3 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(3i)t} + c_2 \overline{v} e^{(-3i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{0t}\big[p\cos(3t) - q\sin(3t)\big],$$
$$r_2 = e^{0t}\big[p\sin(3t) + q\cos(3t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

There is no exponential growth or decay. The real solutions trace closed ellipses around the origin, with period $2\pi/3$; the origin is a center.

Exercise 5.7.12: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -7 & 10 \\ -4 & 5 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 + 2\lambda + 5.$$

A positive-imaginary eigenpair is

$$\lambda = -1 + 2i, \qquad v = \begin{bmatrix} 3 - i \\ 2 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(-1+2i)t} + c_2 \overline{v} e^{(-1-2i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{-1t}\big[p\cos(2t) - q\sin(2t)\big],$$
$$r_2 = e^{-1t}\big[p\sin(2t) + q\cos(2t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

The sine and cosine produce elliptical rotation in the original plane, and the negative real part produces decay. Every nonzero trajectory spirals inward to the origin.

Exercise 5.7.13: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} 4 & -3 \\ 6 & -2 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 - 2\lambda + 10.$$

A positive-imaginary eigenpair is

$$\lambda = 1 + 3i, \qquad v = \begin{bmatrix} 1+i \\ 2 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(1+3i)t} + c_2 \overline{v} e^{(1-3i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{1t}\big[p\cos(3t) - q\sin(3t)\big],$$
$$r_2 = e^{1t}\big[p\sin(3t) + q\cos(3t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

The sine and cosine produce elliptical rotation, while the positive real part expands its size. Nonzero trajectories spiral outward from the origin.

Exercise 5.7.14: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -2 & 1 \\ -8 & 2 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 + 4.$$

A positive-imaginary eigenpair is

$$\lambda = 2i, \qquad v = \begin{bmatrix} 1-i \\ 4 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \qquad q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(2i)t} + c_2 \overline{v} e^{(-2i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{0t}\big[p\cos(2t) - q\sin(2t)\big],$$
$$r_2 = e^{0t}\big[p\sin(2t) + q\cos(2t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

There is no exponential growth or decay. The real solutions trace closed ellipses around the origin, with period $2\pi/2$; the origin is a center.

Exercise 5.7.15: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -8 & -12 & -6 \\ 2 & 1 & 2 \\ 7 & 12 & 5 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = (\lambda - 1)(\lambda + 1)(\lambda + 2).$$

All eigenvalues in this particular computer exercise are real, which is a special case of the requested complex-eigenfunction method. A complete eigenbasis is

$$\lambda_1 = -2, \quad v_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

$$\lambda_2 = -1, \quad v_2 = \begin{bmatrix} -6 \\ 1 \\ 5 \end{bmatrix}.$$

$$\lambda_3 = 1, \quad v_3 = \begin{bmatrix} -4 \\ 1 \\ 4 \end{bmatrix}.$$

Thus the general complex solution is

$$x(t) = \sum_{j=1}^{3} c_j e^{\lambda_j t} v_j, \qquad c_j \in \mathbb{C}.$$

The general real solution is the same formula with $c_j \in \mathbb{R}$, because this basis consists of real vectors. Each eigenpair checks by $Av_j = \lambda_j v_j$, and distinct eigenvalues give a full basis. There are two negative eigenvalues and one positive eigenvalue. The stable plane is $\mathrm{Span}\{v_1, v_2\}$ and the repelling line is $\mathrm{Span}\{v_3\}$. Typical trajectories first combine these modes and eventually move outward in the positive-eigenvalue direction: the origin is a saddle, with no oscillatory spiral in this example.

Exercise 5.7.16: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -6 & -11 & 16 \\ 2 & 5 & -4 \\ -4 & -5 & 10 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = (\lambda - 4)(\lambda - 3)(\lambda - 2).$$

All eigenvalues in this particular computer exercise are real, which is a special case of the requested complex-eigenfunction method. A complete eigenbasis is

$$\lambda_1 = 2, \quad v_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

$$\lambda_2 = 3, \quad v_2 = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}.$$

$$\lambda_3 = 4, \quad v_3 = \begin{bmatrix} 7 \\ -2 \\ 3 \end{bmatrix}.$$

Thus the general complex solution is

$$x(t) = \sum_{j=1}^{3} c_j e^{\lambda_j t} v_j, \qquad c_j \in \mathbb{C}.$$

The general real solution is the same formula with $c_j \in \mathbb{R}$, because this basis consists of real vectors. Each eigenpair checks by $Av_j = \lambda_j v_j$, and distinct eigenvalues give a full basis. All three eigenvalues are positive. The origin is a repeller; generic trajectories eventually align with the fastest growing $\lambda = 4$ eigenvector, while exact invariant eigenspaces retain their own exponential behavior. There is no spiral because the spectrum is real.

Exercise 5.7.17: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} 30 & 64 & 23 \\ -11 & -23 & -9 \\ 6 & 15 & 4 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = (\lambda - 1)\left(\lambda^2 - 10\lambda + 29\right).$$

A positive-imaginary eigenpair is

$$\lambda = 5 + 2i, \qquad v = \begin{bmatrix} 23 - 34i \\ -9 + 14i \\ 3 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 23 \\ -9 \\ 3 \end{bmatrix}, \qquad q = \begin{bmatrix} -34 \\ 14 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(5+2i)t} + c_2 \overline{v} e^{(5-2i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{5t}\big[p\cos(2t) - q\sin(2t)\big],$$

$$r_2 = e^{5t}\big[p\sin(2t) + q\cos(2t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

There is also the real eigenpair $\lambda_0 = 1$, $w = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$. Add $c_0 e^{1t} w$ to the complex solution and $d_0 e^{1t} w$ to the real one. These three independent modes give the entire solution space. The real mode grows like e^t and the spiral-plane modes grow like e^{5t}. Thus the origin is repelling: generic trajectories spiral outward increasingly close to the invariant plane $\mathrm{Span}\{p, q\}$; only the real eigendirection has a nonrotating trajectory.

Exercise 5.7.18: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} 53 & -30 & -2 \\ 90 & -52 & -3 \\ 20 & -10 & 2 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = (\lambda + 7)\left(\lambda^2 - 10\lambda + 26\right).$$

A positive-imaginary eigenpair is

$$\lambda = 5 + i, \qquad v = \begin{bmatrix} 6 + 2i \\ 9 + 3i \\ 10 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 6 \\ 9 \\ 10 \end{bmatrix}, \qquad q = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(5+i)t} + c_2 \overline{v} e^{(5-i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{5t}\big[p\cos(1t) - q\sin(1t)\big],$$

$$r_2 = e^{5t}\big[p\sin(1t) + q\cos(1t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

There is also the real eigenpair $\lambda_0 = -7$, $w = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$. Add $c_0 e^{-7t} w$ to the complex solution and $d_0 e^{-7t} w$ to the real one. These three independent modes give the entire solution space. The line $\mathrm{Span}\{w\}$ is stable, decaying like e^{-7t}. The plane $\mathrm{Span}\{p, q\}$ spirals outward like e^{5t}. Thus this is a spiral saddle: only points on the stable line approach zero, and other trajectories eventually spiral outward.

Exercise 5.7.19: Substitute the circuit constants before solving the voltage system

Example 1 gives

$$A = \begin{bmatrix} -(1/R_1 + 1/R_2)/C_1 & 1/(R_2 C_1) \\ 1/(R_2 C_2) & -1/(R_2 C_2) \end{bmatrix} = \begin{bmatrix} -2 & \frac{3}{4} \\ 1 & -1 \end{bmatrix}.$$

Its eigenvalues are $-\frac{5}{2}, -\frac{1}{2}$, with eigenvector matrix $P = \begin{bmatrix} -3 & 1 \\ 2 & 2 \end{bmatrix}$. Solving $Pc = (4, 4)^T$ gives $c = \begin{bmatrix} -\frac{1}{2} \\ \frac{5}{2} \end{bmatrix}$. Therefore the requested voltage vector is

$$\begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} = \left(-\frac{1}{2}\right) e^{-\frac{5}{2}t} \begin{bmatrix} -3 \\ 2 \end{bmatrix} + \left(\frac{5}{2}\right) e^{-\frac{1}{2}t} \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Each component at zero is the stipulated initial voltage, and differentiating verifies the given circuit equation. Both exponentials decay, consistent with a passive resistive circuit.

Exercise 5.7.20: Substitute the circuit constants before solving the voltage system

Example 1 gives

$$A=\begin{bmatrix}-(1/R_1+1/R_2)/C_1 & 1/(R_2C_1)\\ 1/(R_2C_2) & -1/(R_2C_2)\end{bmatrix}=\begin{bmatrix}-2 & \frac{1}{3}\\ \frac{3}{2} & -\frac{3}{2}\end{bmatrix}.$$

Its eigenvalues are $-\frac{5}{2},-1$, with eigenvector matrix $P=\begin{bmatrix}-2 & 1\\ 3 & 3\end{bmatrix}$. Solving $Pc=(3,3)^T$ gives $c=\begin{bmatrix}-\frac{2}{3}\\ \frac{5}{3}\end{bmatrix}$. Therefore the requested voltage vector is

$$\begin{bmatrix}v_1(t)\\ v_2(t)\end{bmatrix}=\left(-\frac{2}{3}\right)e^{-\frac{5}{2}t}\begin{bmatrix}-2\\ 3\end{bmatrix}+\left(\frac{5}{3}\right)e^{-1t}\begin{bmatrix}1\\ 3\end{bmatrix}.$$

Each component at zero is the stipulated initial voltage, and differentiating verifies the given circuit equation. Both exponentials decay, consistent with a passive resistive circuit.

Exercise 5.7.21: A damped oscillation in current and capacitor voltage

Using Example 3's circuit matrix with the specified constants gives

$$A=\begin{bmatrix}-R_2/L & -1/L\\ 1/C & -1/(R_1C)\end{bmatrix}=\begin{bmatrix}-1 & -8\\ 5 & -5\end{bmatrix}.$$

Its eigenvalues are $-3\pm 6i$. Put $B=A+3I$; direct multiplication gives $B^2=-36I$. Consequently the solution for $x_0=(0,15)^T$ is

$$x(t)=e^{-3t}\big[x_0\cos 6t+(Bx_0/6)\sin 6t\big].$$

Here $Bx_0=(-120,-30)^T$, so

$$i_L(t)=-20e^{-3t}\sin 6t,\qquad v_C(t)=e^{-3t}(15\cos 6t-5\sin 6t).$$

At zero these give 0 amps and 15 volts. Differentiating gives $x'=Ax$; the identity $B^2=-36I$ verifies the formula directly.

Exercise 5.7.22: Solve the second circuit with its stated sign convention

The problem supplies a different circuit matrix; with its constants,

$$A=\begin{bmatrix}0 & 1/L\\ -1/C & -1/(RC)\end{bmatrix}=\begin{bmatrix}0 & 2\\ -2/5 & -4/5\end{bmatrix}.$$

Its eigenvalues are $-2/5\pm(4/5)i$. Set $a=-2/5, b=4/5$ and $B=A-aI$. Then $B^2=-b^2I$ and $Bx_0=(24,-24/5)^T$ for $x_0=(0,12)^T$. The same cosine–sine construction gives

$$i_L(t)=30e^{-2t/5}\sin(4t/5),$$

$$v_C(t)=e^{-2t/5}\big[12\cos(4t/5)-6\sin(4t/5)\big].$$

The initial data and the differential equation both check directly. The signs differ from Exercise 21 because the supplied circuit and current-direction convention differ.